

ECM-Focused Proteomic Analysis of Ear Punch Regeneration in *Acomys cahirinus*

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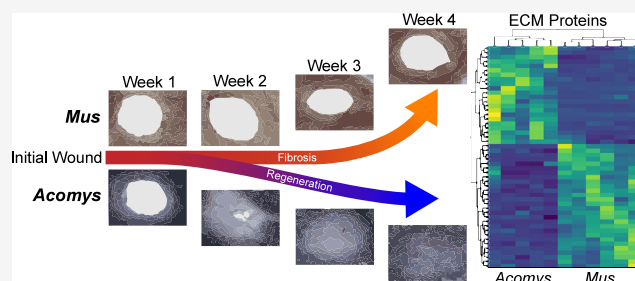
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ABSTRACT: While significant acute injury leads to a functionally deficient fibrotic scar in mammals, the spiny mouse *Acomys cahirinus* has evolved scar-free regeneration of severely damaged tissue. Extracellular matrix (ECM) composition is a critical regulator of wound repair, and previous studies have suggested that alterations in its expression may be responsible for the differences in regenerative capacity observed between *Mus musculus* and *A. cahirinus*, yet analysis of this critical tissue component has been limited in previous studies by its insolubility. Here, we utilize a two-step ECM-optimized extraction to perform proteomic analysis of tissue composition during wound repair after full-thickness ear punches in *A. cahirinus* and *M. musculus* from weeks 1 to 4 postinjury. We observe changes in various ECM proteins implicated in wound regeneration and fibrosis, including collagens, provisional matrix, coagulation proteins, and matricryptic signaling domains. We additionally report differences in cross-linking enzyme activity and ECM protein solubility between species. Furthermore, we observed rapid and sustained increases in pro-regenerative macrophage markers in *Acomys* but not in *Mus*. Together, these findings indicate that ECM differences contribute to the pro-regenerative restoration of functional tissue seen in *Acomys* and implicate potential targets for future therapies to attenuate scar formation in other mammals.

KEYWORDS: extracellular matrix, fibrosis, wound repair, regeneration, collagen, macrophage, *Acomys*



INTRODUCTION

In mammals, tissue damage repair is mediated through a series of events including inflammatory response, granulation tissue formation, and extracellular matrix (ECM) remodeling.¹ During acute wound healing in most mammalian tissues, ECM remodeling typically results in the replacement of healthy tissue with fibrotic scar tissue² which is characterized by excessive deposition of irregularly organized structural ECM proteins, primarily fibrillar collagen.³ Alternatively, insufficient ECM remodeling leads to altered matrix-regulated autophagy and results in chronic wounds or ulcers.⁴ Additionally, ECM composition has been shown to regulate a number of key wound healing processes, including inflammation,^{5,6} fibroblast activity,⁷ immune cell polarization,⁸ and tissue-specific cell differentiation.⁹ Given its role in tissue functionality, ECM-focused regenerative medicine research, which aims to utilize these regulatory effects to restore healthy tissue after damage occurs, has emerged as a prominent focus of study.

In contrast to most mammals, the spiny mouse (genus: *Acomys*) demonstrates full restoration of tissue architecture and function following substantial acute injury. *Acomys* displays dermal properties that include brittle skin with low tensile strength and high regenerative capacity, which appear to have evolved as a mechanisms to evade predators in the wild.¹⁰

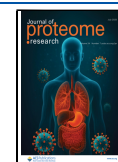
Because tissue strength is dictated in large part by its ECM protein composition,¹¹ it is likely that there are large-scale differences in matrix composition between the skin of *Acomys* and that of *Mus musculus*, which has much stronger skin and lacks the same regenerative capacity. While other examples of regeneration in mammals have been identified (e.g., liver regeneration,¹² embryonic skin¹³), *Acomys* represents the first demonstration of full skin autotomy in mammals.^{10,14,15} Subsequent studies have demonstrated that this capacity for nonfibrotic repair additionally extends to a variety of internal tissues, including spinal cord,^{16,17} myocardium,^{18–20} skeletal muscle,^{21,22} and kidney.²³ Outside of its regenerative ability, *A. cahirinus* was already in limited use as a laboratory organism due to its other unique biological similarities to humans including propensity for diabetes,²⁴ menstruation,²⁵ production of steroids in the adrenal glands,²⁶ and nephrogenesis before birth.²⁷ Because of these features, *A. cahirinus* has become a

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commonly used nonmodel organism for cross-species comparison in wound healing.

Due to its diverse composition containing skin, hair follicles, adipose, muscle, and cartilage, ear punch tissue is a useful model for complex tissue regeneration. Following ear punch injury, *Acomys* has been found to rapidly close the resulting wound and regenerate all tissue types.^{10,14} In contrast, *Mus musculus* (C57BL/6) heal their wounds through fibrotic scarring with little to no evidence of tissue regeneration. While *M. musculus* ear punches showed no significant closure and appeared fully healed by day 14 postinjury, *Acomys* injuries displayed a newly generated ring of translucent tissue which continued to grow inward until 100% of ear punches were fully closed by day 56.¹⁴

Previous proteomic studies investigating ear punch regeneration have been conducted in *Acomys*,¹⁵ however, numerous experimental limitations allow for significant improvements. First, earlier studies performed MS/MS ion searches of *Acomys* proteomic data using the *M. musculus* proteome database, preventing identification of *Acomys* peptide sequences which are nonhomologous to their *Mus* counterparts and subsequently limiting the comparisons between *Mus* and *Acomys* that can be performed. Second, previous data suggests large-scale differences in ECM composition between *Mus* and *Acomys*, which may be partly responsible for *Acomys* regenerative capacity.^{10,15} However, due to difficulties in solubilizing the ECM via traditional proteomic methods,²⁸ like those employed in previous studies of ear punch regeneration,¹⁵ a comprehensive analysis of the ECM proteome in *Acomys* is lacking. Consequently, a deeper, more thorough ECM-focused proteomic approach can provide novel insights into the regenerative properties of *Acomys*. Third, previous proteomic analyses of ear punch regeneration involved samples collected only up to day 14 postinjury.¹⁵ Other studies have shown that >85% of wound closure and new tissue growth occurs after day 14 in *Acomys*,¹⁴ indicating that continued sampling after day 14 is essential for capturing the proteome of newly synthesized tissue.

Here, we performed ECM-optimized 2-step extraction for proteomic analysis via LC-MS/MS on the healing edge of full-thickness 4 mm ear punch wounds from *Mus* and *Acomys* sampled weekly from 0 to 4 weeks postinjury. The use of hydroxylamine chemical digestion, as employed in this study, to extract insoluble ECM proteins has been demonstrated to provide improved identification of ECM components over standard proteomic methods in previous studies across a variety of sample types.^{28–30} Additionally, we utilize previously published RNA-sequencing data³¹ to generate an annotated *Acomys* protein sequence database for interpretation of the acquired of the mass spectrometric data. These methods enable more comprehensive and accurate characterization of the ECM across *Mus* and *Acomys*, allowing for improved comparison of matrix signatures between species and the generation of novel insights regarding candidate drivers of scarless tissue regeneration in this unique mammalian model.

MATERIALS AND METHODS

Acomys cahirinus Protein Sequence Database Generation

An *A. cahirinus* specific protein database was constructed from a previously published transcriptome³¹ (Tr2aacds_v2.fasta). The TransDecoder.LongOrfs tool in Transdecoder v5.5.0³² (github.com/TransDecoder/TransDecoder) was used to extract open reading frames from the transcriptome and translate to putative

proteins. The resulting protein sequences were searched against the Uniprot database³³ (release 2021_03) using BLASTp v2.12.0³⁴ with flags -max_target_seqs 1, -outfmt 6, and -evalue 1e-5. TransDecoder.Predict was then run using results from the BLASTp search to filter out spurious proteins with no homology to known proteins. The resulting putative protein set was filtered to retain only complete proteins (those including both a start and stop codon). The rmdup tool in seqkit v2.0.0³⁵ was used to identify instances where multiple ORFs were predicted to produce identical proteins, for which a single ORF was retained.

To annotate putative proteins, we used a reciprocal best BLAST (RBB) followed by one-way best BLAST (OBB) approach with the *Mus musculus* protein database (version GRCm39; NCBI: GCA_000001635.9) as the subject database. In instances where a *Mus* protein was assigned an RBB hit in our protein set, all other OBB hits to that *Mus* protein were removed. In instances where multiple proteins had OBB, but not RBB, hits to the same *Mus* protein, the protein with the highest bitscore was retained as the best match and all others removed. This approach resulted in a database of 18,682 putative *Acomys* proteins with homology to *Mus* proteins.

Sample Acquisition

The Seattle Children's Research Institute's (SCRI) Institutional Animal Care and Use Committee (IACUC) approved all animal procedures. Adult male and female *Mus musculus* (The Jackson Laboratory, CD1(ICR)) and *Acomys cahirinus* (colony at SCRI) were maintained within the Seattle Children's Research Institute's onsite vivarium. CD1 mice were housed in a pathogen-free room maintained on 12:12 (Light:Dark) lighting schedule, *Acomys* were housed in a separate room maintained on 14:10 (Light:Dark) schedule, and all animals received food and water ad libitum. *Acomys* and *Mus* adult males and females were anaesthetized with 4% (v/v) vaporized isoflurane (Henry Schein Animal Health, Dublin, OH), and a 2 mm thumb punch (Kent Scientific) was used to generate punches across the medial ear pinna in the right and left ears. Initial punches were performed across 24 individual mice and collected as uninjured tissue samples (n = 6 used for study). Punches were then collected from the healing edge of the wound at 7, 14, 21, and 28 days (1 to 4 weeks) postinjury, collecting at only a single postinjury time point for each original injury site. All samples were immediately flash frozen in liquid nitrogen after collection for storage.

Sample Preparation

Dissected tissue samples were received frozen and lyophilized prior to analysis. Briefly, lyophilization was performed by inserting frozen samples into a Clear Complete Fast-Freeze flask (Labconco) and immediately placing under vacuum (<100 μ Bar) using a FreeZone 4.5L benchtop freeze-dryer (Labconco, catalog number: 7750020). Lyophilized samples were cut into <1 mm² pieces using a razor blade and weighed before extraction. Samples were processed using a 2-step extraction beginning with 3 successive washes with decellularization buffer (50 mM Tris-HCl (pH 7.4), 0.25% CHAPS, 25 mM EDTA, 3 M NaCl). Approximately 100 mg of 3 mm glass beads were used to mechanically agitate samples in a Bullet Blender (NextAdvance) prior to all extraction steps. For decellularization, samples were vortexed at 4 °C for 20 min after agitation, centrifuged at 18,000 xg for 20 min, and the supernatant was collected. Successive decellularization washes were pooled by sample for analysis. After decellularization, sample pellets were subjected to chemical solubilization with 1 M hydroxylamine hydrochloride (HA) in 4.2 M Gnd-HCl, 0.2 M K₂CO₃, pH 9.0 at 42 °C for 4 h

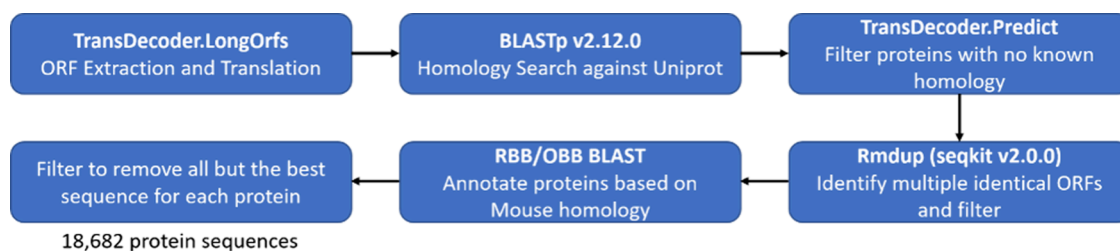


Figure 1. Workflow for annotated protein database generation from RNA-seq data.

with constant agitation at 120 rpm on an orbital shaker (Labnet Orbit M60). After digestion, samples were centrifuged at 18,000 $\times g$ for 20 min and the supernatant was collected. This 2-step extraction produced separate cellular and ECM fractions for each sample. Extracted fractions were precipitated using ice-cold 80% acetone (Fisher Scientific #A929–1), washed 2 $\times$ with ice-cold 100% acetone, and brought up in 8 M urea prior to analysis. Protein concentration of each fraction was measured using A660 Protein Assay (Pierce). Proteolytic digestion of sample extracts was carried out according to the FASP protocol³⁶ using 10 kDa molecular weight cutoff filters (Sartorius Vivacon 500 #VN01H02) and loading 10 μg of protein resulting from each fraction. Samples were prepared by reducing with 10 mM dithiothreitol (DTT) for 30 min at room temperature (RT, 25 $^{\circ}C$), alkylating with 22 mM iodoacetamide (IAM) for 15 min at RT, and digesting with trypsin (1:100) at 37 $^{\circ}C$ for 14 h. Peptides were recovered from the filter using successive washes with 0.2% formic acid (FA).

LC-MS/MS Analysis

Digested peptides (200 ng) were loaded onto individual Evotips and desalted following the manufacturers protocol and separated on an Evosep One chromatography system (Evosep, Odense, Denmark) using the default 30 samples per day LC method over a Pepsep column (150 μm inter diameter, 15 cm) packed with ReproSil C18 1.9 μm , 120 $\text{\AA}$ resin. The system was coupled to the timsTOF Pro mass spectrometer (Bruker Daltonics, Bremen, Germany) via the nanoelectrospray ion source (Captive Spray, Bruker Daltonics). The mass spectrometer was operated in PASEF mode. The ramp time was set to 100 ms and 10 PASEF MS/MS scans per topN acquisition cycle were acquired. MS and MS/MS spectra were recorded from m/z 100 to 1700. The ion mobility was scanned from 0.7 to 1.50 Vs/cm². Precursors for data-dependent acquisition were isolated within ± 1 Th and fragmented with an ion mobility-dependent collision energy, which was linearly increased from 20 to 59 eV in positive mode. Low-abundance precursor ions with an intensity above a threshold of 500 counts but below a target value of 20000 counts were repeatedly scheduled and otherwise dynamically excluded for 0.4 min.

Data Analysis

Raw data was searched using MSFragger (version 3.5) through the FragPipe platform (version 18.0). Precursor tolerance was set to ± 15 ppm and fragment tolerance was set to ± 0.08 Da, allowing for 2 missed cleavages. *M. musculus* data was searched against SwissProt (17,029 sequences) restricted to *M. musculus*, while *A. cahirinus* data was searched against the custom sequence database described above (18,682 sequences). To identify proteins not included in the *Acomys* database, *A. cahirinus* data was additionally searched against SwissProt restricted to *M. musculus*. The only protein reported in the manuscript using this analysis approach is lysyl oxidase (LOX), denoted with “*” in

Figure 6. All databases had common contaminants added using version 1.1 of the CRAPome.³⁷ Trypsin semispecific cleavage was used in all searches. Fixed modifications were set as carbamidomethyl (C). Variable modifications were set as oxidation (M), oxidation (P) (hydroxyproline), Gln $\rightarrow$ pyro-Glu (N-term Q), deamidated (NQ), and acetyl (protein N-term). Results were filtered to 1% false discovery rate (FDR) at the peptide and protein level using Philosopher (version 4.4.0) employed through FragPipe. Label-free quantification (LFQ) was performed using the IonQuant (version 1.8.0), including unique and razor peptides. Cell and ECM fractions for each organism were searched independently and data was normalized to total LFQ intensity by fraction, so that total signal in each fraction was equal across all samples before fractions were aggregated for analysis. Protein entries in the *Acomys* database covering less than 50% of the mouse sequence length were not utilized for comparisons or included in normalization. For *Acomys* data searched using the *Mus* database, peptides were filtered to include only those with a quantified identification in at least one sample from both *Acomys* and *Mus*, removing peptides which correspond to a *Mus* sequence that does not match the *Acomys* sequence, before applying total intensity normalization by fraction as described above. Reported core matrisome (ECM proteins) and matrisome associated proteins were annotated with MatrisomeDB.³⁸ Results of database searching and subsequent processing are tabulated in tables S1 and S2.

Statistical Analysis

To determine statistical significance in protein expression levels between species, data underwent statistical comparison in R software (4.4.2) with tidyverse (2.0.0), matrixTests (0.2.3), circlize (0.4.16), and lme4 (1.1–35.5) packages. Summed total differences were compared using Student's *t* test. Time-dependent changes in protein levels were tested using repeated measures 2-way ANOVA to identify significantly different protein expression trajectories between species. P-values were corrected for multiple comparisons using False Discovery Rate correction, and statistical differences were determined to be $q < 0.05$. Full results of statistical analysis can be found in tables S3 (mixed modeling) and S4 (*t* test). For chord diagrams (Figures 8 and S2), partial correlation coefficients were calculated from Spearman's correlation rho between the residuals of linear models of each protein accounting for time, and the top 5 partial correlation values for each cell marker were displayed. Chord diagram plots were generated using the CIRCOS visualization tool.³⁹

Protein solubility (Figure 6) was calculated as the normalized intensity for each protein in the ECM fraction divided by the total normalized intensity for that protein (sum of intensities from cell and ECM fractions). Student's *t* test was performed to determine statistical significance of differences in solubility between uninjured samples in *Mus* and *Acomys*, as well as

between W0 and W4 samples for *Acomys* and *Mus* individually. Student's *t* test was additionally performed to determine significance of differences in collagen signaling peptide abundance (Figure 5).

RESULTS AND DISCUSSION

Acomys cahirinus Protein Sequence Database Generation

To improve identification of *Acomys* peptides nonhomologous to *Mus* sequences, we first generated a protein sequence database (DB) using previously published RNA-sequencing (RNA-seq) data³¹ (Figure 1). In brief, open reading frames were extracted and translated from RNA-seq data before performing a homology search against UniProt with unrestricted taxonomy. Multiple filtering steps were performed to remove aberrant transcripts before annotating protein sequence based on their homology to *M. musculus* protein entries. Finally, the database was filtered to remove redundant sequences for the same protein entry, resulting in a single protein sequence per gene (18,682 protein sequences) (Figure 1).

To assess the quality of our newly generated *Acomys* protein database, a comparison of *Acomys* MS data searched against both the *M. musculus* SwissProt database (17,268 sequences), and in-house *A. cahirinus* (18,682 sequences) database was performed (Table 1). The *A. cahirinus* DB resulted in a 47% increase in total

Table 1. Comparison of *Acomys* and *Mus* Database Performance for *Acomys* Data Analysis

	<i>Acomys</i> vs <i>Mus</i> DB(previous approach)	<i>Acomys</i> vs <i>Acomys</i> DB(current approach)	<i>Mus</i> vs <i>Mus</i> DB(gold standard)
Total Peptides Identified	53566	78951	77863
ECM Peptides Identified	5850	11495	10512
Total Proteins Identified	4652	4794	6154
ECM Proteins Identified	252	264	330

peptides and nearly a 2-fold increase in ECM peptides, demonstrating significant improvement in peptide IDs across the proteome. Additionally, *Acomys* data searched against the in-house DB provides an additional 1,088 peptide sequences and 983 ECM peptide sequences compared to *Mus* data searched against the *Mus* database. However, while the *Acomys*-specific DB results in 142 additional protein identifications for *Acomys* data compared to the *Mus* DB, the best-performing *Acomys* data analysis results in only 67% of total and 80% of ECM *Mus* protein identifications (Table 1). To assess detectable matrisome proteins which are missing from the *Acomys* DB, the contents of the *Acomys* DB were compared to the full list of matrisome proteins identified in our previous study of 25 mouse organs using 3-fraction ECM-focused proteomics.⁴⁰ Matrisome proteins which were identified in this large-scale ECM study but are missing from the *Acomys* DB described here (114 proteins) are listed in table S7.

Animal Model and ECM Proteomics Study Design. This study utilized the model of ear punch regeneration employed on *Acomys* and *Mus* as described by Brewer et al.⁴¹ followed by ECM-focused proteomic analysis (Figure 2). In brief, 2 mm

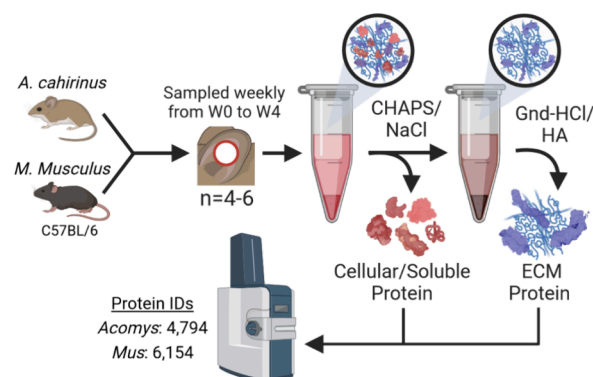


Figure 2. Workflow for proteomic analysis of ear punch regeneration in *A. cahirinus* and *M. musculus* (C57BL/6). Full-thickness 4 mm ear punches were performed on *A. cahirinus* and *M. musculus* ($n = 4-6$ per time point) and the excised tissue was collected (W0). Tissue was then collected from the healing edge of the ear punch wounds weekly from week 1 (W1) to week 4 (W4). Samples were subjected to a 2-step ECM-optimized extraction, digested, and analyzed using a timsTOF Pro MS system.

punches were taken from the central region of *A. cahirinus* and *M. musculus* ears ($n = 24$). At weeks 1 to 4 post injury, the healing edge of the wound was excised, collecting the healing edge at only one time point from each animal ($n = 4-6$ per time point). Samples were then subjected to a 2-step ECM-focused protein extraction method, consisting of decellularization using CHAPS and 3 M NaCl followed by ECM extraction using hydroxylamine-HCl (HA) and guanidine-HCl (Gnd-HCl).²⁸⁻³⁰ Extracted fractions were analyzed via LC-MS/MS, resulting in 4,794 protein IDs for *Acomys* and 6,154 protein IDs for *Mus*.

Baseline Differences in ECM Composition. To establish baseline differences between the proteomes of *Mus* and *Acomys*, comparisons were performed between initial, uninjured ear punch samples using data from cell and ECM fractions (Figure 3). Significant differences were observed in a wide range of cellular proteins, including cytoskeletal components (actins, tubulins, and cytoskeletal keratins) and metabolic enzymes (Figure 3A). While not the primary focus of this study, cellular protein alterations between uninjured *Mus* and *Acomys* tissue indicate baseline differences in metabolic capacity and cellular structure between the two species.

Large-scale differences in matrisome protein composition were also observed between uninjured *Mus* and *Acomys* (Figure 3B, S1). We observed elevated abundance of fibrillar collagens COL1A2 and COL5A1 in *Mus*, alongside collagen-supporting proteoglycan decorin (DCN), indicating a more robust fibrillar collagen network in *Mus* which likely contributes to the greater tensile strength of *Mus* tissue. Of the collagens most upregulated in *Acomys*, including COL4A5, COL5A3, COL6A3, COL9A1, COL9A3, COL11A2, COL12A1, COL18A1, and COL27A1, only two are fibrillar (COL5A3 and COL27A1), with the rest represented by FACIT, network, and other collagens. The unique expression profile of minor collagens in *Acomys* may contribute to its altered tissue strength and regenerative capacity.⁴² Differences in noncollagen matrix proteins were also detected, including elevated galectin-3 (LGALS3) and chondroadherin (CHAD) in *Mus* and elevated matrilin-2 (MATN2), and laminin chains in *Acomys* (Figure 3B). Many of these differentially regulated ECM proteins have been implicated in either pro-regenerative or pro-inflammatory processes during wound repair (Table 2).

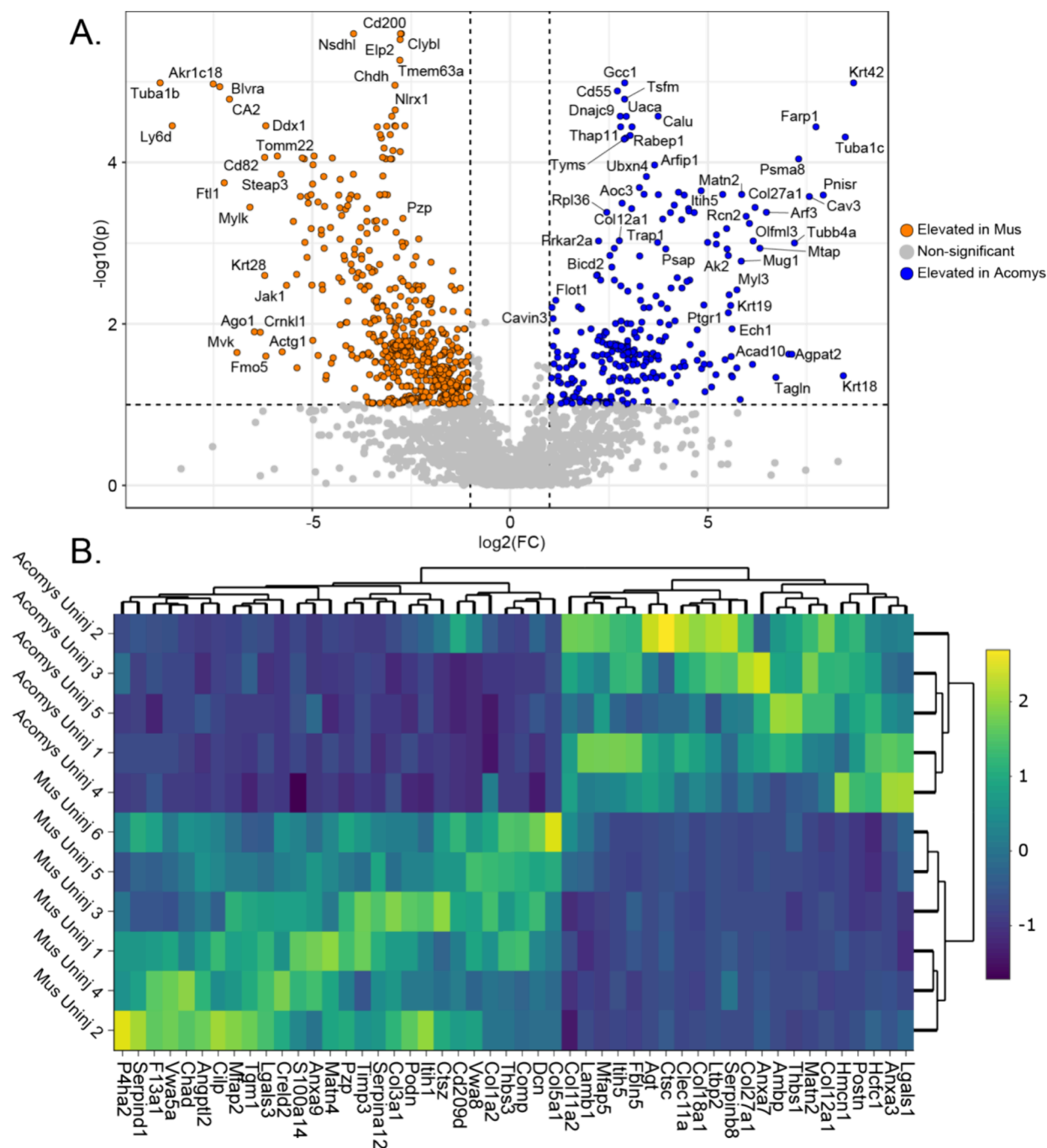


Figure 3. Baseline comparison of uninjured *Mus* and *Acomys*. A) Volcano plots showing all proteins significantly elevated in uninjured (W0) *Mus* (left, orange) and *Acomys* (right, blue). B) Heatmap displaying top 50 distinguishing ECM proteins by *t* test between uninjured *Mus* and *Acomys*.

Time Series Analysis of ECM Composition During Wound Repair. Analysis of ECM protein expression during the course of wound healing in *Acomys* and *Mus* reveals functionally significant clusters of differentially expressed proteins between organisms (Figure 4). Enriched gene ontology terms from each protein cluster can be found in figure S2. Cluster 1 reveals proteins which are expressed at moderate to low intensity in *Acomys* and high intensity in *Mus*, with many proteins stably expressed and some decreasing in abundance over the course of wound healing. These proteins include fibrillar collagens COL1A2, COL3A1, and COL5A1, all of which have been associated with and are overexpressed in fibrosis and scar formation.² We also observe cornified envelope protein

S100A14 as well as transglutaminase 1 (TGM1), the principal enzyme responsible for cross-linking of the cornified envelope,⁴³ in this cluster.

Cluster 2 shows proteins elevated in *Mus* compared to *Acomys*. However, in this cluster we observe induction of many proteins after injury in both species, with intensity remaining high throughout the course of wound healing in *Mus* but not in *Acomys*. Proteins assigned to this cluster include a variety of proteases involved in ECM remodeling (cathepsin A (CTSA), cathepsin H (CTSH), disintegrin and metalloproteinase domain-containing protein 9 (ADAM9)) as well as ECM protease inhibitors (Interleukin-1 trypsin inhibitor heavy chain H2 (ITIH2), corticosteroid-binding globulin (SERPINA6), leuko-

Table 2. Selected Differentially Expressed Proteins Enriched in *Acomys* (Top) and *Mus* (Bottom) with Identified Roles in Wound Healing and Regeneration

Gene	<i>Acomys</i> Expression	<i>Mus</i> Expression	Wound Healing Role	KO Effect
POSTN	High	Low, moderate induction after injury	Regulates COL fibrillogenesis and skin viscoelasticity ⁵¹ /Associated with myofibroblast transition ⁵²	Impaired wound closure, ⁵² reepithelialization, ⁵³ and COL cross-linking ⁵¹
FBLN2	Low in uninjured, strong induction after injury	Low, weak induction after injury	Maintains basement membrane integrity, ^{54,55} participates in formation of provisional FN1 matrix ⁵⁶	Skin blistering, reduced basement membrane integrity ⁵⁵
MATN2	High, low at week 1 with later recovery	Very low in uninjured, moderate induction after injury	Multiahesive adaptor protein essential for the regeneration of muscle, nerve, skin, and other tissues ⁵⁷	Myoblast cell cycle arrest, delayed muscle regeneration ⁵⁸
AEBP1	High	Low in uninjured, moderate induction at week 2	Regulate fibroblast proliferation, COL production and MSC differentiation ⁵⁹	Delayed wound repair correlating with defects in fibroblast proliferation ⁵⁹
THBS1	High, increased at weeks 2 and 3	Low, slight induction after injury	Activates TGF- β , regulates angiogenesis ⁶⁰ and COL assembly, mediates ER stress response ⁶¹	Delayed repair, loosely compacted and disorganized granulation tissue ⁶⁰
HSPG2	High in uninjured, lower at week 1 with some recovery	Moderate in uninjured, lower after injury	Orchestrates morphogen signaling, ⁴⁷ stabilizes ECM, promotes angiogenesis ⁴⁸	Delayed wound healing in deficient mice ⁶²
C1QC	Low in uninjured, strong induction after injury	Low in uninjured, moderate induction after injury	Supports new vessel formation, promotes wound healing as DDR2 ligand ⁶³	Defective vessel formation, lower vessel density, impaired wound healing ⁶⁴
PODN	Moderate	High, increased at week 4	Negative regulator of cell migration and proliferation ⁶⁵	Increased smooth muscle cell proliferation ⁶⁶
TGM1	Moderate	High	Cross-links cornified envelope ⁴³	Impaired skin barrier function ⁶⁷
SDC1	Not detected in uninjured, slight induction after injury	Low in uninjured, strong induction after injury	Modulates integrin activation, upregulated during fibrosis ⁶⁸	Delayed wound healing ⁶⁸
CHAD	Low, weak induction after injury	High in uninjured, Moderate after injury	Binds collagen and inhibits fibrillogenesis in vitro, reduces chondrocyte differentiation ⁶⁹	Increased chondrocyte differentiation and collagen II deposition ⁶⁹
ADAM9	Very low	Low in uninjured, strong induction after injury	Causes collagen XVII ectodomain shedding ⁷⁰	Accelerated wound repair and keratinocyte migration ⁴⁴
COL14A1	Low	Low in uninjured, strong induction at week 2	Regulation of collagen fibrillogenesis ⁷¹	Reduced tensile strength of tissue ⁷¹

cyte elastase inhibitor A (SERPINB1A), heparin cofactor 2 (SERPIND1)). Of note, ADAM9 knockout in mice has been shown to accelerate cutaneous wound healing.⁴⁴ Additionally, prolyl hydroxylases A1 and A2 (P4HA1 and P4HA2) are observed in this cluster and are involved in the generation of essential post-translational modifications (PTMs) during collagen synthesis. The persistence of enzymes related to ECM remodeling and synthesis over the course of wound healing in *Mus* but not *Acomys* suggests a longer duration of ECM deposition and remodeling after injury in *Mus*. Clusters 3 and 4 represent proteins which are expressed at high levels in both species throughout the course of wound repair. Proteins in cluster 3 are generally not significantly different between species and include a variety of collagen, laminin, and annexin proteins. Proteins in cluster 4 display slightly higher average intensity in *Acomys* than in *Mus*, including COL4A2, COL5A2, and cross-linking enzymes transglutaminase 2 (TGM2) and lysyl oxidase homologue 1 (LOXL1), in addition to a variety of serpin protease inhibitors.

Cluster 5 displays proteins which are significantly elevated in *Acomys* compared to *Mus*, with generally larger induction after injury in *Acomys* (Figure 4). This cluster contains a number of proteins which have been associated with a pro-regenerative phenotype or whose absence has been implicated in impaired wound healing in previous studies, including periostin (POSTN), fibulin-2 (FBLN2), matrilin-2 (MATN2), adipocyte enhancer-binding protein 1 (AEBP1), and thrombospondin-1 (THBS1) (Table 2). GO term enrichment analysis indicates that biological process terms “regulation of cell migration”, “growth”, and “angiogenesis” are significantly enriched among proteins in cluster 5 (Figure S2E). We additionally observe a number of proteins related to coagulation (fibrinogen alpha chain (FGA), plasminogen (PLG), prothrombin (F2)), early wound healing, and provisional matrix deposition (fibronectin (FN1)) in this cluster, suggesting a

more robust early wound healing response in *Acomys* compared to *Mus*. FN1 plays a key role in the formation of early provisional matrix, and FBN2 has previously been shown to colocalize and coexpress with FN1 during early stage of skin regeneration, forming a critical component of the provisional FN1 matrix.⁴⁵

Cluster 6 contains proteins which are detected at lower levels at week 4 postinjury than in uninjured tissue in both species, with generally higher expression in *Acomys* throughout the course of wound repair. These proteins include basement membrane components nidogen-1 (NID1) and perlecan (HSPG2), both of which are essential for the construction and maintenance of the basement membrane. NID1 deficiency during skin wound healing was found to induce hyperproliferation and differentiation delays as well as aberrant laminin deposition and distribution, demonstrating a critical role of NID1 during tissue repair.⁴⁶ HSPG2, as well as being a critical structural component of the basement membrane, regulates the binding of morphogens and mitogens to cells,⁴⁷ altering signaling pathways and cell behavior, assists with collagen fibrillogenesis,⁴⁷ and promotes angiogenesis.⁴⁸ HSPG2 is highly conserved across multicellular organisms and has been found to be critical for wound repair in the *Nematostella vectensis* model of regeneration.⁴⁹

In cluster 7, we observe proteins which are present at low levels in uninjured samples in both species. In *Acomys*, all proteins contained in this cluster show significant upregulation after injury coupled with recovery toward uninjured levels after week 1 postinjury, while in *Mus* upregulation is observed in only a subset of these proteins. All proteins in this cluster are identified at the highest intensity in *Acomys* week 1 postinjury samples except for tenascin-C (TNC), which is identified at the greatest intensity in *Acomys* week 2 postinjury. Proteins in this cluster are highly enriched in components of the coagulation cascade, including fibrinogen beta chain (FGB), fibrinogen gamma chain (FGG), coagulation factor XIII B chain (F13B),

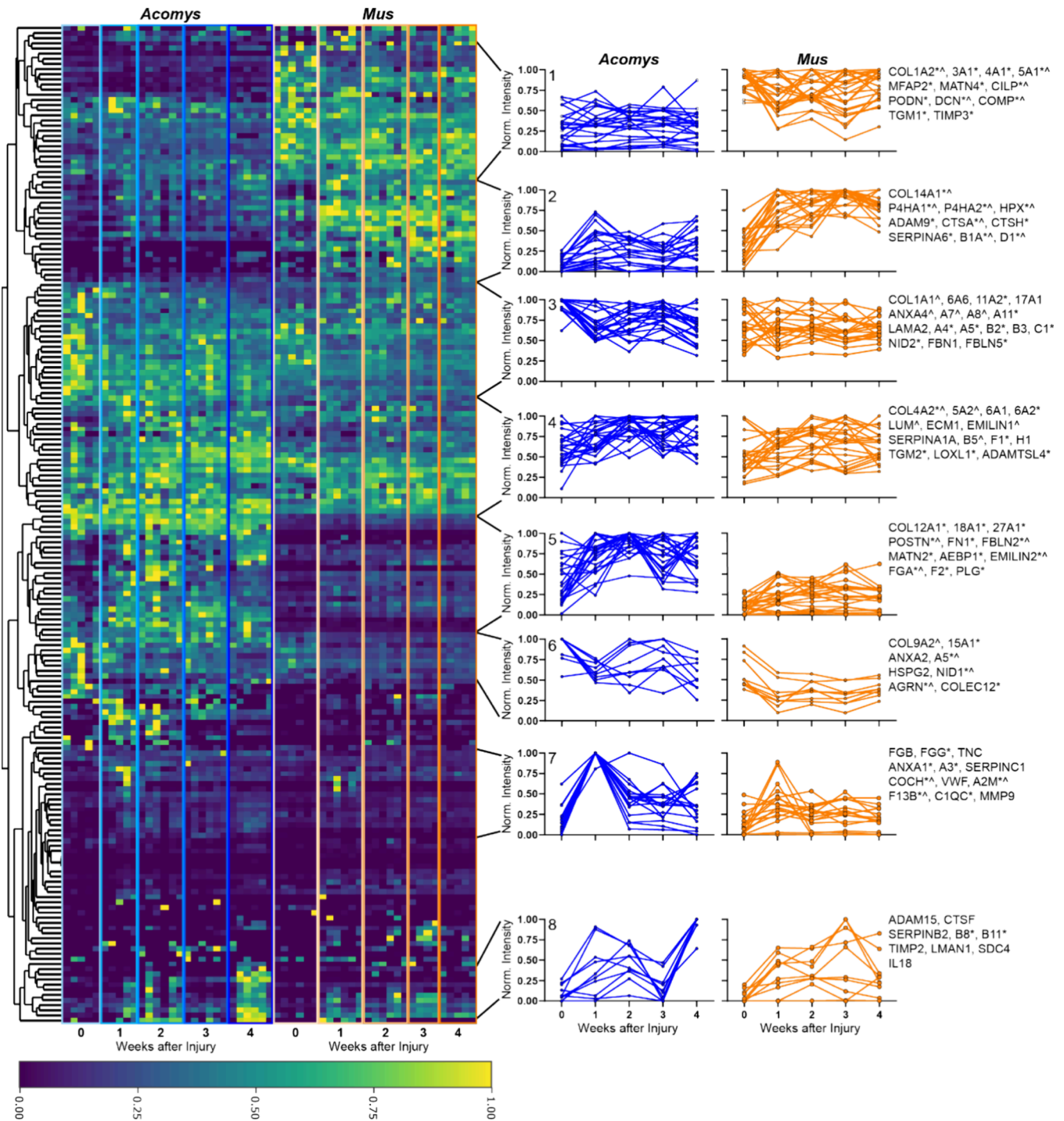


Figure 4. Expression profiles of matrisome proteins throughout wound healing reveal functionally significant expression clusters. Heatmap displaying normalized intensity for all core matrisome proteins across *Acomys* (blue) and *Mus* (orange) samples from uninjured (0 weeks after injury) to week 4 (left). Profile plots displaying average intensity (normalized to maximum intensity for each protein across all samples) from uninjured to week 4 postinjury in *Acomys* (blue) and *Mus* (orange) for all identified proteins in the specified cluster (right). Cluster numbers are indicated in the upper-left corner of *Acomys* profile plots. Significantly different proteins between species via *t* test across all time points are labeled with “*”. Proteins which significantly vary between species in their abundance at later time points compared to baseline (determined via linear mixed modeling) are labeled with “^”.

cochlin (COCH), and von Willebrand factor (VWF). The increased expression of a wide variety of coagulation factors at week 1 post injury in *Acomys* compared to *Mus* demonstrates a more significant coagulation response in *Acomys* which persists for at least 1 week. This upregulation of coagulation cascade proteins may contribute to a more robust early wound healing

response, representing the formation of a critical initial matrix which stabilizes wound fields, promotes cell migration and proliferation, and ultimately sets the stage for effective regeneration to occur.⁵⁰ We additionally observe significant upregulation of TNC in *Acomys* compared to *Mus* which persists after coagulation proteins have dissipated. GO term analysis

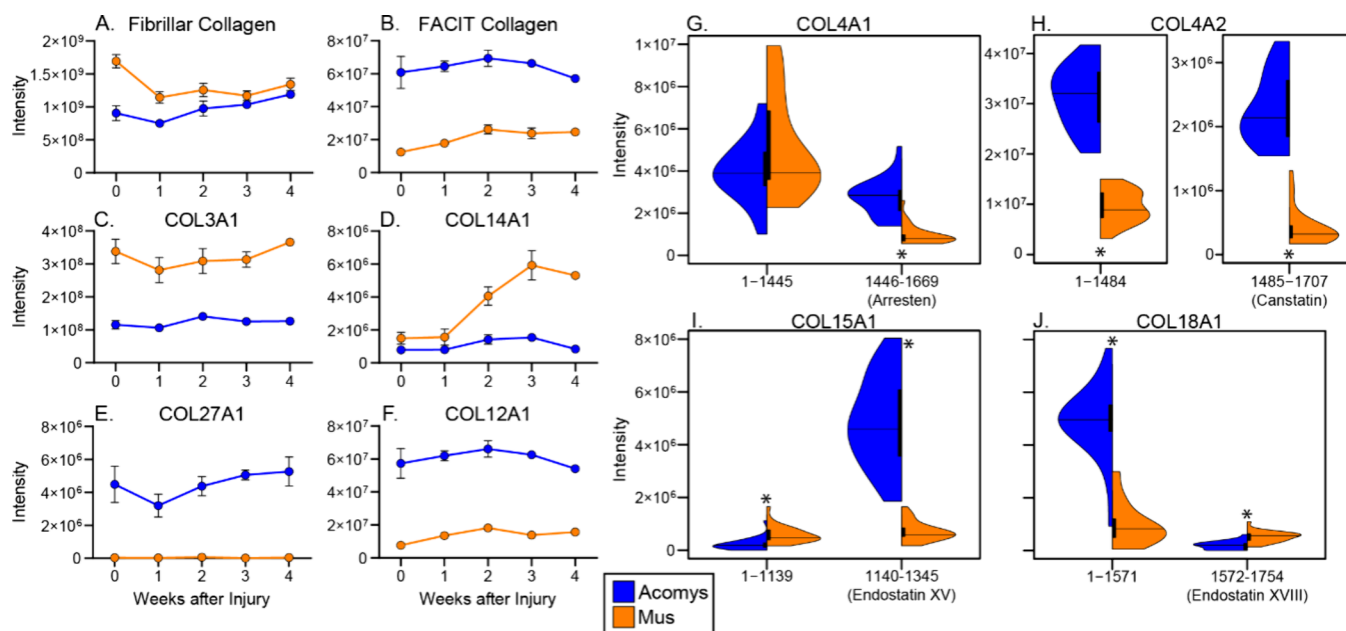


Figure 5. Collagen composition during wound healing varies between *Acomys* and *Mus* and collagen signaling peptides are regulated independently from their parent proteins. Summed intensity for fibrillar collagens (A), FACIT collagens (B), COL3A1 (C), COL14A1 (D), COL27A1 (E), and COL12A1 (F) in *Acomys* (blue) and *Mus* (orange) across the course of wound healing. Uninjured samples are labeled as 0 weeks after injury. G) Intensity for COL4A1 residues 1–1445 (left) and residues 1446–1669 (arresten, right, $p < 0.001$) between *Acomys* and *Mus* across all samples. H) Intensity for COL4A2 residues 1–1484 (left, $p < 0.001$) and residues 1485–1707 (canstatin, right, $p < 0.001$). I) Intensity for COL15A1 residues 1–1139 (left, $p < 0.001$) and residues 1140–1345 (endostatin XV, right, $p < 0.001$). J) Intensity for COL18A1 residues 1–1571 (left, $p < 0.001$) and residues 1572–1754 (endostatin XVIII, right, $p < 0.001$). Significant comparisons in G–J ($p < 0.05$) indicated by "*", horizontal lines represent median value.

reveals that biological process terms "response to stress", "defense response", and "blood coagulation" are significantly enriched in this pathway, indicating that *Acomys* presents a more robust initial defense response compared to *Mus* (Figure S2G).

Cluster 8 includes proteins that are elevated at later time points compared to uninjured tissue in both *Acomys* and *Mus*. In general, proteins in cluster 8 display earlier induction in *Acomys* compared to *Mus*, with most proteins detected at maximum abundance at weeks 1–2 postinjury in *Acomys* and at week 3 postinjury in *Mus*. Proteins in this cluster are primarily represented by ECM modifiers and their regulators, including disintegrin and metalloproteinase domain 15 (ADAM15), serpin protease inhibitors, and tissue inhibitor of metalloproteinase 2 (TIMP2). Alterations in ECM regulator expression demonstrate varied ECM remodeling between species throughout the course of wound repair, with remodeler induction occurring later after injury in *Mus* than in *Acomys*.

Collagen's Contribution to Wound Repair. Collagens represent the primary structural component of the ECM and play a critical role in both regeneration and the development of fibrosis after injury. Analysis of collagen abundance during wound healing reveals many significant differences in collagen subtype expression between *Acomys* and *Mus* (Figure 5). Scar formation is typically characterized by the deposition of disordered fibrillar collagen.⁷² We observe significantly greater abundance of fibrillar collagen in *Mus* compared to *Acomys* in uninjured samples as well as weeks 1 and 2 postinjury. However, increased deposition of fibrillar collagen during scar formation is unexpectedly not observed in *Mus* (Figure 5A). This trend does not hold true for all fibrillar collagens, with fibrillar collagens COL3A1 and COL27A1 displaying expression patterns which are distinct from one another as well as from that of all fibrillar

collagens combined (Figure 5C, E). While in other models of scarless healing, such as fetal tissue regeneration, COL3A1 is consistently reported to be expressed at higher levels in tissue which performs scarless regeneration than in fibrotic tissue,^{1,73,74} we observe elevated COL3A1 in *Mus* compared to *Acomys* (Figure 5C). Like other fibrillar collagens, COL3A1 contributes significantly to tissue tensile strength and knockouts or mutations of the gene in mouse skin cause thin and fragile skin which undergoes increased wounding.⁷⁵ The relative lack of COL3A1 in *Acomys* likely contributes to reduced skin strength and suggests that *Acomys* regeneration occurs through mechanisms independent of collagen fibril diameter regulation by COL3A1. Fibrillar collagen COL27A1, on the other hand, is detected at great abundance in *Acomys* at all time points (Figure 5E). COL27A1 is unique from other fibrillar collagens due to its short, interrupted main helical domain⁷⁶ and, due to its relatively recent discovery,⁷⁷ little is known about its function in the context of wound healing. Studies have shown that COL27A1 is expressed at high levels in developing cartilage⁷⁷ and its expression is regulated by SRY-Box transcription factor 9, a key regulator of chondrogenesis.⁷⁸ Thus, elevated COL27A1 in *Acomys* likely indicates greater representation of this protein in uninjured tissue and improved regeneration of cartilage after injury.

Fibrillar-associated collagens with interrupted triple helices (FACIT collagens), which bind the exterior of collagen fibrils to regulate fibrillogenesis and confer various biological functions, display the opposite trend, with strong enrichment in *Acomys* throughout the analyzed time course (Figure 5B). In contrast with this pattern, induction of FACIT collagen COL14A1 is observed after week 1 postinjury in *Mus* but not in *Acomys*. COL14A1 has been shown to contribute to tissue resilience⁷¹

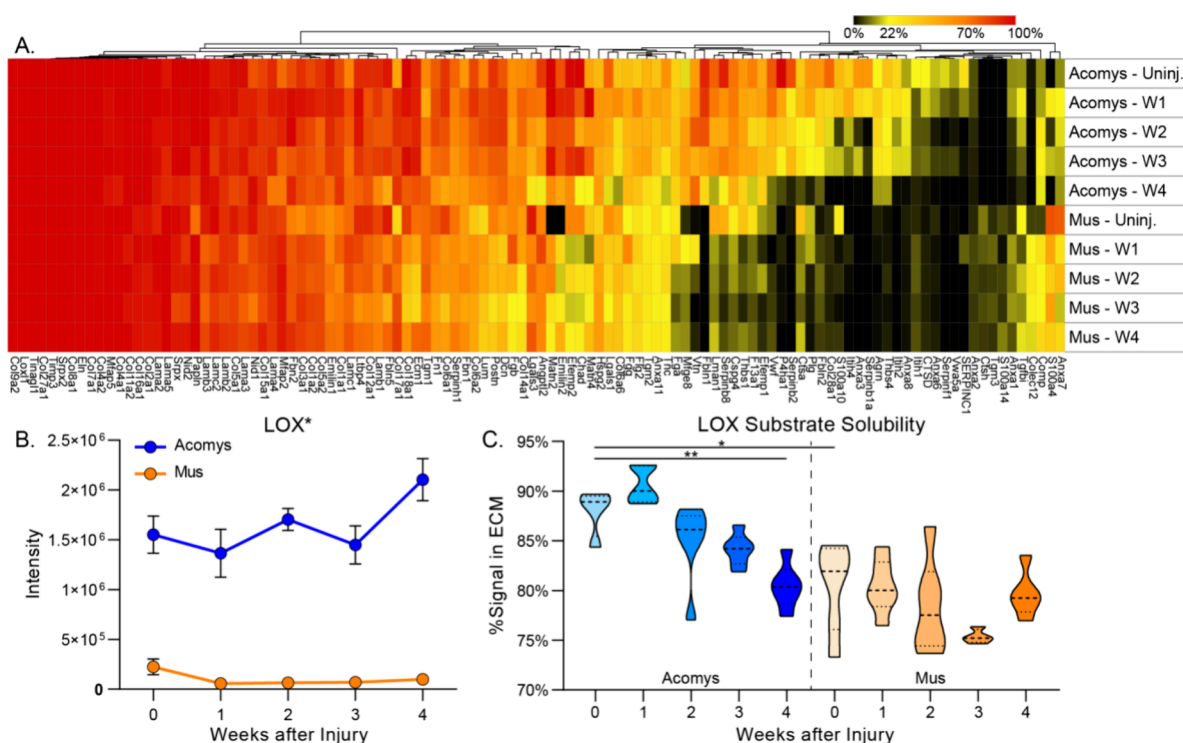


Figure 6. Solubility of LOX substrates and other ECM proteins varies by species and time point. A) Heatmap displaying solubility of ECM proteins measured by the percentage of total signal identified in the ECM fraction for each protein. B) Abundance of LOX in *Mus* and *Acomys* throughout the course of wound healing. *Intensity for *Acomys* samples was derived from searches against the *Mus* database using only shared peptides between species. C) Average solubility of LOX substrates in *Mus* and *Acomys* over the course of wound healing. All identified collagen chains and elastin were included as LOX substrates for this analysis.

and is expressed at high levels during porcine fibroproliferative scarring after deep cutaneous injury in pigs,⁷⁹ indicating that it may play a role in *Mus* scar formation as well. COL12A1, a FACIT collagen which binds type I collagen fibrils to regulate fiber assembly and adhere other ECM proteins to the fibrillar matrix,⁸⁰ is detected at significantly higher abundance in *Acomys* than in *Mus* throughout the course of wound healing (Figure 5F). Knockout of COL12A1 has been shown to delay wound healing through failed sequestration of TGF- β , while over-expression delays healing through increased persistence of M1 macrophages.⁸⁰

Enrichment of Regenerative Matrikines in *Acomys*. The C-terminal noncollagenous domains of type IV, XV, and XVIII collagens can function independently from the structural roles of their parents proteins as potent inhibitors of angiogenesis, motility, and other pro-proliferative cell behaviors.^{81,82} COL4A1 and COL4A2 form the primary structural components of the basement membrane and contain noncollagenous signaling domains termed arresten⁸³ and canstatin,⁸⁴ respectively. Both arresten and canstatin have been shown to inhibit tube formation and proliferation in many types of endothelial cells,⁸¹ reduce both PI3K/Akt activation and vascular endothelial growth factor (VEGF) expression,⁸⁵ and induce endothelial cell apoptosis in certain contexts.^{84,86} Previous studies have utilized mass spectrometry-based approaches to identify changes in matrikine abundance in models of tissue aging.⁸⁷ While we observe no difference in median intensity between *Acomys* and *Mus* for the primary collagenous region of COL4A1 (residues 1–1445), we detect significantly elevated abundance of peptides corresponding to arresten (residues 1446–1669) (Figure 5G). In COL4A2, on the other hand, we

observe significant elevation in *Acomys* of both canstatin (residues 1485–1707) and the remainder of the parent protein, with no evident difference in expression pattern between the two regions (Figure 5H). Thus, elevated abundance of the canstatin peptide is likely explained by increased expression of the full COL4A2 chain, while arresten may represent a matrikine-specific effect. Peptide data for collagens described in Figure 5G–J can be found in table S5.

The $\alpha 1$ chains of type XV and XVIII collagens can also be cleaved to produce molecules termed endostatins which primarily function to inhibit angiogenesis.^{88,89} However, key differences have been identified between endostatin-XV (derived from COL15A1) and endostatin-XVIII (derived from COL18A1) which drive the distinct biological functions reported for these highly related proteins.⁹⁰ Endostatins XV and XVIII vary significantly in their affinity for common ECM ligands and endostatin-XV lacks the zinc and heparin binding ability which is present in endostatin XVIII and required for inhibition of FGF-2-induced angiogenesis.⁹¹ Additionally, endostatins vary in their localization within tissue structures: endostatin-XVIII is expressed more within elastic fibers and forms a distinct component of portal veins while endostatin XV is a much more prominent component in muscle tissue.⁹¹ We observe greater intensity for the collagenous region of COL15A1 (residues 1–1139) in *Mus*, contrasting with a strong enrichment of endostatin-XV (residues 1140–1345) in *Acomys* (Figure 5I). Endostatin-XVIII displays the opposite trend, showing significant elevation in *Mus* while its parent protein is strongly enriched in *Acomys* (Figure 5J).

While numerous studies have been performed analyzing the effects of endostatin-XVIII in various contexts, including wound

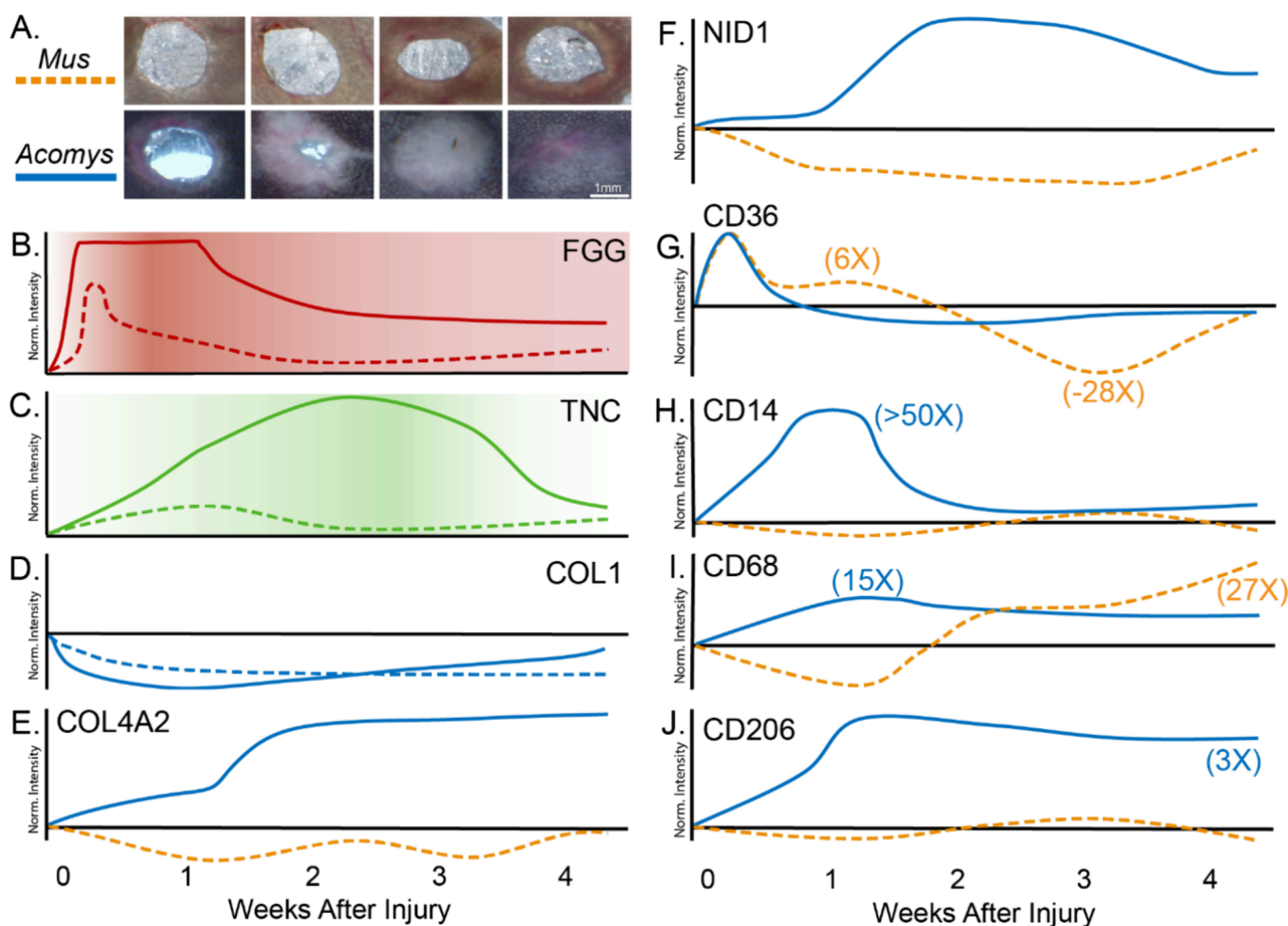


Figure 7. Wound healing trajectory for *Acomys* and *Mus*. A) Images of ear punch injuries at 1-, 2-, 3-, and 4-weeks postinjury. Full wound closure and depigmentation is demonstrated in *Acomys* samples. (B–J) Line plots display ECM proteins and cell markers over the course of wound repair, represented as intensity normalized to change over baseline (*Acomys*, solid line; *Mus*, dashed line). ECM markers include FG (B), TNC (C), COL1 (D), COL4A2 (E), and NID1 (F). Cell markers include CD36 (platelet, G), CD14 (monocyte, H), CD68 (monocyte/macrophages, I), and CD206 (M2 macrophages, J). Numbers in parentheses represent fold change over baseline at the corresponding time point. Images in Figure 7A are reproduced with permission from Brewer et al.⁴¹ Copyright 2021 Elsevier.

healing, much less is known about the independent *in vivo* function of endostatin-XV.⁹⁰ Endostatin-XVIII overexpression has been shown to result in delayed wound healing,⁹² while deposition of full-length COL18A1 has been demonstrated in the basement membrane during re-epithelialization of mouse skin wounds.⁹³ The dual role of COL18A1 during skin regeneration is mirrored in our data, where in *Acomys* the parent COL18A1 protein is present at much higher levels than the C-terminal endostatin fragment (Figure 5J). This expression pattern likely allows for COL18A1 to participate in basement membrane formation without inducing the antiproliferative effects caused by endostatin-XVIII. Further studies are required to elucidate the role of endostatin-XV in *Acomys* regenerative capacity.

Protein Solubility and Cross-linking Enzyme Comparison. Analysis of protein solubility between species over the course of wound healing reveals two significant trends: (1) generally decreased protein solubility in uninjured tissue compared to late wound-healing time points and (2) lower solubility in *Acomys* compared to *Mus*, especially in uninjured tissue. Lysyl oxidase (LOX) is responsible for cross-linking of collagen and elastin fibrils, decreasing their solubility in the process.^{94,95} In *Acomys*, LOX significantly increased in

abundance at every analyzed time point (Figure 6B), which was coupled with significantly reduced solubility in *Acomys* compared to *Mus* in uninjured ($p = 0.016$), week 1 ($p = 0.00006$), week 2 ($p = 0.027$), and week 3 ($p = 0.00008$) time points (Figure 6C). Collagen cross-linking is usually associated with tissue strength;^{96,97} however, we observed higher levels of LOX in *Acomys* where skin tensile strength is very low. We hypothesize that while altered expression of collagen subtypes likely contributes to reduced tensile strength of *Acomys* skin (Figure 5), increased cross-linking may help maintain tissue integrity and architecture during normal physical activity. Additionally, multiple studies have shown that up-regulation of LOX during the early stages of wound healing can result in accelerated wound healing.^{98,99} Insufficient levels of LOX during granulation tissue formation can lead to scarring⁹⁷ and LOX has been shown to play an important role in the regulation of angiogenesis,^{100,101} an important process for supplying the granulation matrix with nutrients necessary for regeneration.¹⁰²

While LOX-family enzyme activity is typically increased during scar formation, both hypertrophic and keloid scar tissue actually display less enzymatic collagen cross-linking than does healthy tissue or mature scar.⁷² Similarly, in a model of postburn scarring, LOX was found at low activity in granulation tissue

with a sharp increase after 2–3 months, with elevated activity maintained in the tissue for 3–5 years.¹⁰³ It is possible that our methods were unable to capture the elevation in LOX activity and expression which occurs at later wound healing time points during scar formation in *Mus*.

In addition to a general trend of decreased solubility at later time points in many ECM proteins (Figure 6A), we observed a significant reduction in LOX substrate solubility in *Acomys* week 4 samples compared to uninjured samples (Figure 6C). The presence of a significant solubility difference between uninjured and week 4 tissue in *Acomys* but not in *Mus* likely demonstrates the increased production of immature, incompletely cross-linked matrix during *Acomys* wound healing. Interestingly, in *Acomys* we detect LOX at the highest abundance in week 4, when protein solubility is highest. This increased LOX abundance after full wound closure may be responsible for cross-linking and regenerating the tensile strength of newly synthesized tissue in the wound area. Previous studies have shown that although 100% of *Acomys* ear punches closed by day 56 postinjury, cartilage architecture remains histologically immature at this point, with morphologically mature cartilage observed at 3 months postinjury.¹⁴ LOX cross-linking activity plays a key role in continued tissue remodeling after wound closure,⁹⁷ with the effects of this activity on solubility accumulating over time as cross-links are formed. It is likely that analysis of later time points would reveal recovery toward uninjured solubility levels as tissue maturation continues.

Differential Patterns of Cell Marker Expression.

Analysis of changes from baseline expression for key ECM components and cell markers throughout the wound healing trajectories of *Acomys* and *Mus* reveals important insights about how wound repair differs in these two models (Figure 7). Notably, *Acomys* exhibited a more rapid, pronounced, and prolonged increase in Fibrinogen gamma chain (FGG) levels (Figure 7B), suggesting an enhanced clotting response, supported by similar patterns in other clotting proteins (FGA and FGB). As previously mentioned, increased fibrin deposition in *Acomys* may represent the formation of a critical initial matrix which aids in effective tissue regeneration.⁵⁰ TNC is a key member of the early provisional matrix formation during wound healing. We found that *Acomys* displayed a more robust induction of TNC after week 1 postinjury compared to *Mus*, where TNC induction was much less pronounced (Figure 7C). This significant difference in TNC expression suggests that *Acomys* may establish a more substantial provisional matrix, which has been shown to play a key role in fetal scarless wound repair⁷⁴ and to promote regeneration in a variety of other contexts.¹⁰⁴ Type 1 collagen is a primary structural component of scar tissue and is generally increased during scar formation. Interestingly, we observed a greater recovery of COL1 levels toward baseline after injury in *Acomys* compared to *Mus* (Figure 7D). This may indicate that *Acomys* possess a heightened ability to remodel and resolve scar tissue, potentially contributing to their regenerative advantage.

We additionally observe increased postinjury induction of basement membrane proteins in *Acomys* but not in *Mus* (Figure 7E, F). COL4A2 and NID1, crucial structural components of the basement membrane, both display slight elevation over baseline at week 1 and strong induction by week 2 postinjury in *Acomys*. In *Mus*, on the other hand, these proteins display lower levels at week 1 compared to baseline with some recovery back to baseline levels by week 4. Basement membrane deposition is a

critical element of skin repair and is consistent with wound re-epithelialization.¹⁰⁵

Analysis of cell marker expression also reveals differing dynamics of cellular infiltration between *Mus* and *Acomys* during wound repair. CD36, a platelet marker, was induced in both *Acomys* and *Mus* following injury (Figure 7G). However, the platelet population remained near baseline levels through week 4 in *Acomys*, while a drastic drop in platelet marker signal was observed at week 3 in *Mus*. This differential response suggests that *Acomys* may maintain a more sustained platelet-mediated wound healing response, which could be beneficial for efficient clot formation and tissue repair. Alternatively, rapid and sustained angiogenesis in the newly regenerated tissue in *Acomys* may capture circulating platelets at the time points sampled. CD14, a monocyte marker, displayed a remarkable difference in induction between the two species (Figure 7H). *Acomys* exhibited over a 50-fold induction over baseline at week 1, which subsequently dropped back to baseline levels at week 2. In contrast, *Mus* showed little to no induction over baseline at any time point (Figure 7H). Additionally, CD68, a pan-macrophage marker, exhibited a 15-fold increase above baseline at week 1 in *Acomys*, which remained elevated throughout the 4-week course (Figure 7I). In contrast, *Mus* displayed a drop in expression below baseline at week 1, followed by a substantial increase to 27-fold above baseline at week 4 (Figure 7I). These findings suggest that *Acomys* experience greater initial macrophage infiltration at week 1, while *Mus* exhibit delayed but persistent macrophage induction, potentially reflecting the persistence of an inflammatory environment in *Mus* during later stages of wound repair. Markers of monocyte/macrophage infiltration indicate a greater and more rapid influx of monocytes and macrophages into the wound site after injury in *Acomys* compared to *Mus*, likely contributing to their enhanced regenerative capacity.

While cell markers indicate greater macrophage contribution to early wound repair in *Acomys*, all macrophages are not created equal in the context of wound healing. M1 macrophages are primarily associated with the early inflammatory response, producing pro-inflammatory cytokines and helping to clear pathogens and debris.¹⁰⁶ In contrast, M2 macrophages are pro-regenerative, secreting anti-inflammatory cytokines and promoting tissue repair by stimulating cell proliferation and angiogenesis.¹⁰⁷ The presence of M2 macrophages, often marked by CD206 expression, is a crucial determinant of effective wound healing and tissue regeneration.¹⁰⁸ In our study, CD206 exhibited significant induction in *Acomys*, with a 3-fold increase over baseline at week 1 which persisted until week 4 (Figure 7J). In stark contrast, no induction of CD206 was observed at any time point in *Mus*, implying greater and more persistent infiltration of pro-regenerative M2 macrophages after injury in *Acomys* (Figure 7J). The sustained presence of M2 macrophages in *Acomys* may play a pivotal role in tissue repair, dampening inflammation, and promoting regenerative processes throughout the wound healing continuum.^{107,109} Additionally, the long pentraxin PTX3, a pattern recognition molecule and essential component of humoral innate immunity, has been shown to play a critical role in the orchestration of tissue repair, with PTX3 deficiencies resulting in increased fibrin and collagen deposition during wound repair.¹¹⁰ In addition to its primary immune function, PTX3 regulates wound healing through a variety of mechanisms,¹¹¹ including interaction with plasminogen to drive remodeling of the fibrin-rich inflammatory matrix¹¹² and resolution of inflammation via macrophage modulation.¹¹³

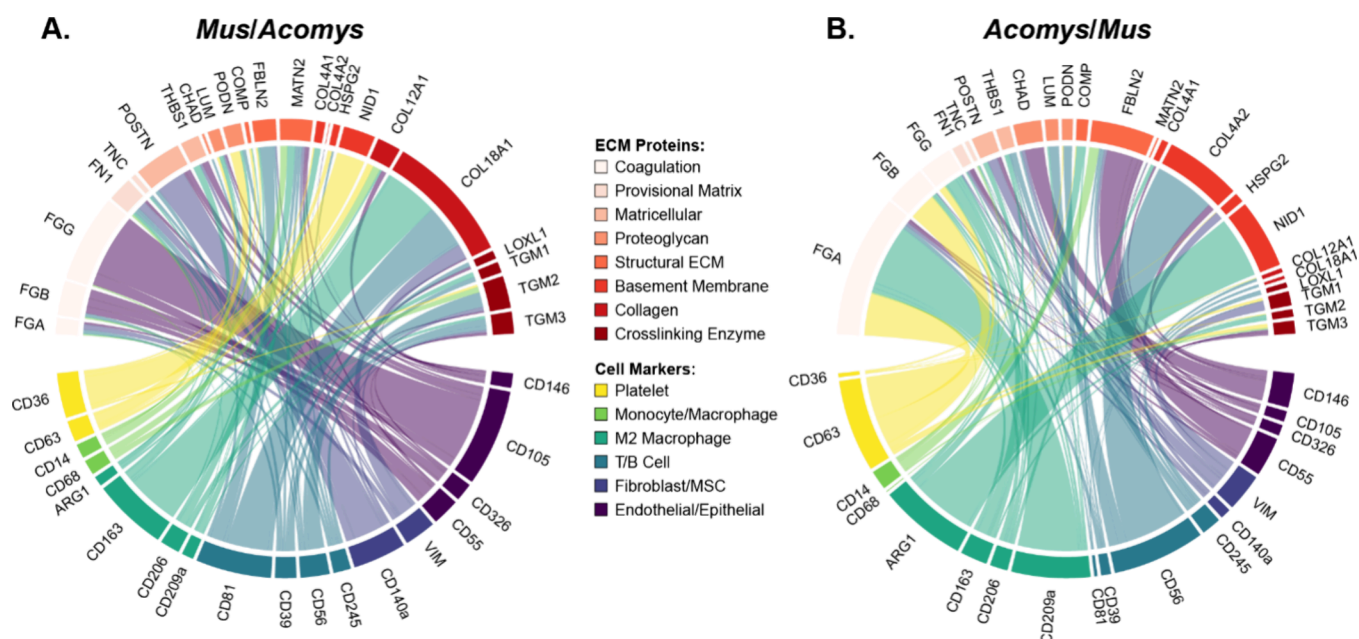


Figure 8. Chord diagrams of *Mus* vs *Acomys* correlations between cell markers and ECM protein pairs. A) Strength of associations between cell markers and ECM protein pairs determined by partial correlation coefficient ratios of *Mus*/*Acomys*. B) Strength of associations between cell markers and ECM protein pairs determined by partial correlation coefficient ratios of *Acomys*/*Mus*. Line thickness indicates relative correlation value ratio.

The presence of PTX3 was shown to cause reductions in a variety of pro-inflammatory macrophage markers, including IL-1 β , TNF- α , and CD86, demonstrating the ability of this protein to repress pro-inflammatory M1 macrophage differentiation.¹¹³ In our data, we observe strong induction of PTX3 at week 4 in *Acomys*, while all other sample groups display low or undetectable abundance (data not shown). Elevated levels of PTX3 in *Acomys* at later time points may aid in the resolution of macrophage-induced inflammation and promote pro-regenerative immune cell activity. These findings provide valuable insights into the biological mechanisms driving the regenerative capacity of *Acomys*, with implications for understanding wound repair and regeneration in other species.

To identify potential relationships between immune cell infiltration and the resulting composition of ECM in newly generated tissue, we performed correlation analysis and generated chord diagrams using the CIRCOS visualization tool³⁹ to represent correlation ratios between *Mus* and *Acomys* (Figure 8). Individual correlations between cell markers and ECM proteins for *Mus* and *Acomys* can be found in figure S3. The distinct patterns of ECM component and cellular marker correlations between *Acomys* and *Mus* provide a window into their differential wound healing and regenerative capacities. In *Mus*, there is a notable induction of endothelial and epithelial markers (CD105, CD326) in response to injury, which correlates with early expression of coagulation proteins (FGB, FGG). Specifically, CD105, an endothelial marker, shows induction in *Mus* at week 1 that is maintained through week 4, coupled with sparse identification in *Acomys*. This suggests early endothelial activation in *Mus* which is consistent with fibrotic wound repair via VEGF signaling.¹¹⁴ CD326, an epithelial marker, exhibits more pronounced induction at weeks 1–3 in *Mus* compared to *Acomys*, indicating a higher baseline epithelial presence that may contribute to the more fibrotic phenotype observed in *Mus*.

Conversely, in *Acomys*, the correlations between M2 macrophage markers and coagulation proteins are more pronounced,

indicating different immune cell dynamics during early wound repair. CD209a, an M2 macrophage marker, shows strong induction at week 1 after injury in *Acomys*, in contrast to the very low levels displayed in *Mus* throughout the measured time course. The coincident expression of coagulation proteins and pro-regenerative macrophage markers indicates a more robust early anti-inflammatory response in *Acomys*. Greater correlations of the platelet marker CD63 to fibrinogen component proteins, with induction postinjury and maintenance at higher levels until week 4, further underscores the more robust coagulation response which may contribute to regenerative capacity in *Acomys*.

CD163 (a marker of M2 macrophages) and CD81 (a marker of T and B cells) display much greater correlations to COL18A1 in *Mus* than in *Acomys*. Unlike in other cases, where greater correlation is driven by greater expression in that species, here the correlation between these proteins is driven by their low-level, stable expression in *Mus* compared to their more dynamic and varied expression across the time course in *Acomys*. As previously discussed, COL18A1 plays a role in basement membrane repair while M2 macrophages are key contributors to a pro-regenerative inflammatory environment. Together, the postinjury dynamic expression of these markers indicates a more active immune environment in *Acomys*, coupled with greater synthesis of minor structural components for tissue repair.

The correlation between ARG1, another M2 macrophage marker, and NID1, a basement membrane protein, is greater in *Acomys* than in *Mus*, which may indicate a maintained interaction between M2 macrophages and the basement membrane in *Acomys*, rather than platelets. This coincident expression contrasts with the pattern in *Mus*, where ARG1 is induced at week 1 but not correlated with NID1, suggesting a disconnection in the cellular and ECM dynamics essential for regeneration. However, in contrast to other markers of M2 macrophages, ARG1 is detected at higher levels in *Mus* than in *Acomys*, likely due to the nonspecific expression of this

important metabolic enzyme in other cell types as well as M2 macrophages.^{115,116}

The natural killer T-cell marker CD56 displays a much greater correlation to basement membrane collagen COL4A2 in *Acomys* than in *Mus*. This correlation difference is a result of greater postinjury induction of both proteins in *Acomys* than in *Mus*, indicating more significant infiltration of natural killer T-cells at week 1 postinjury and concurrent basement membrane synthesis during early wound repair. Basement membrane repair is a critical step in the regeneration of functional tissue.¹¹⁷ The proteoglycan lumican (LUM) also displays greater correlation to CD56 and postinjury induction in *Acomys* than in *Mus*, and LUM deficiency has been shown to cause defects in collagen fiber organization.¹¹⁸ Increased basement membrane repair and regulation of collagen fibrillogenesis likely contribute to the superior regenerative capabilities observed in *Acomys*.

In summary, these findings suggest that *Acomys* has evolved a sophisticated ECM and cellular response that facilitates regeneration through modulation of immune responses, early deposition of basement membrane components, and restoration of ECM structure after injury. This contrasts with the response in *Mus*, where a more static and fibrosis-prone ECM environment is observed. These insights into the regenerative biology of *Acomys* provide clues for understanding how dynamic reciprocity between cell populations and ECM composition influences regenerative capacity.

Comparison with Other Regenerative Models. The study of regenerative processes in mammals has tremendous potential to generate insights which result in improved clinical wound repair. In addition to *Acomys*, other models have also contributed to our understanding of scar-free wound healing, including the axolotl (*Ambystoma mexicanum*),¹¹⁹ fetal mammals,^{120–123} the MRL/MpJ mouse,¹²⁴ rabbit ear skin,^{125,126} and reindeer antler,^{127,128} among others. However, limited proteomics work has been done on these models and most previous studies have inadequately represented ECM proteins. By comparing protein expression patterns identified during *Acomys* regeneration to those observed in other regenerative models, we can potentially identify tissue components which drive regeneration across a variety of mammalian contexts. Data for all proteins discussed in these comparisons can be found in table S6.

The capacity for scar-free regeneration of adult wounds is generally much less common in mammals than it is in other vertebrates,^{129,130} appearing in fish,^{131,132} amphibians,¹³³ and salamanders¹³⁴ which have been extensively studied over the past 50 years.¹³⁵ One such nonmammalian vertebrate, the axolotl, is able to fully regenerate entire limbs as well as a variety of internal organs, displaying capacity for regeneration which is comparable to that of *Acomys* and greater than that of either frogs or zebrafish.¹¹⁹ The incredible regenerative ability of both *Acomys* and axolotls provides a unique opportunity for identification of ECM components which contribute to their shared regenerative capacity. A recent review compared expression of ECM-related genes between *Acomys* and axolotl and identified a number of shared expression patterns that may contribute to regeneration in these models, including shared upregulation of COL12A1, TNC, FN1, THBS1, THBS2, and C1QC between the two species.¹³⁶ Many proteins reported as upregulated in both *Acomys* and axolotl display concurrent upregulation in our data, demonstrating strong agreement between our study and previous analyses. However, analyses incorporated into this review were limited to transcriptomics

and analysis of the soluble ECM fraction,¹³⁶ limiting the conclusions that can be drawn regarding additional alterations to the insoluble, fibrillar ECM that we are able to capture in this study.

Reindeer antlers are an intriguing model of regeneration due to their rapid growth and extensive remodeling, representing the only mammalian organ that completely regenerates on an annual basis.¹³⁷ Independent analysis of both the *Acomys* and reindeer antler models reveals certain protein expression patterns which are shared between the two models during regeneration. COL12A1 is elevated in the regenerating regions of both models, while fibrillar collagens COL1A2 and COL3A1 are reduced both in antler growth centers and in *Acomys* compared to *Mus* after injury.¹²⁷ COL12A1 has also been detected at high levels in zebrafish during heart¹³⁸ and spinal cord¹³¹ regeneration, suggesting a role for this protein in regeneration across species and tissues. In addition to observed shared expression patterns, reindeer antler and *Acomys* also exhibited differences in ECM composition which may indicate variation in regenerative mechanism between these two models. For example, thrombospondin 4 (THBS4) was found to be elevated in the reindeer antler growth center but is reduced in *Acomys* healing tissue compared to *Mus* in this analysis. THBS4 is involved in promoting fibroblast migration and angiogenesis, displaying upregulation during wound repair in *Mus* and *H. sapiens*.¹³⁹

A similar injury model to the one used in this paper has been employed since the 1970s to study regeneration of full-thickness ear punches in rabbit ears. In this model, full-thickness ear punches up to 1 cm in diameter are repaired by the regeneration of new tissue following blastema formation. Comparison of rabbit ear regeneration with *Acomys* ear regeneration has revealed a potential role for endostatin in regulating angiogenesis to improve wound regeneration. Both systemic¹⁴⁰ and local¹⁴¹ injections of Endostar, a recombinant form of human endostatin-XVIII, have been shown to reduce hypertrophic scar formation in rabbit ears. Interestingly, elevated levels of endostatin-XV, but not endostatin-XIII, were observed in *Acomys* during wound regeneration, independent of the parent protein (Figure S1, J).

Finally, a comparison of fetal scarless wound healing to ear punch regeneration in African spiny mice has revealed some interesting insights into the role of specific proteins in regeneration. In comparison to adult tissue, earlier and more intense deposition of cellular adhesion molecules, including fibronectin (FN1) and tenascin-C (TNC), has been reported to drive fibroblast recruitment and re-epithelialization in human fetal skin repair. We observe elevated levels of both FN1 and TNC in *Acomys* compared to *Mus* at early wound healing time points. Additionally, decorin (DCN), which is involved in the regulation of collagen fibrillogenesis as well as the binding and storage of growth factors within the ECM, is displayed at low levels in injured fetal skin in contrast to injured adult skin. In our data, DCN was observed to have greater expression in *Mus* compared to *Acomys*. While we observe ~20% greater signal for integrins overall in *Mus*, some integrins, including ITGAV and ITGA5, are specifically upregulated in *Acomys* at week 1. ITGAV and ITGA5 are regulators of adipocyte differentiation, where elevated expression reduces differentiation and increases maintenance of the stem cell population.

Overall, these comparative studies provide important insights into the complex interplay of ECM proteins in scar-free wound healing and open new avenues for future research. Further

investigation is needed to fully understand the mechanisms underlying these differences and to determine how they can be harnessed to improve wound healing in humans.

CONCLUSIONS

The research presented here was undertaken to improve understanding of mammalian regeneration through analysis of differences in ECM composition between *Acomys* and *Mus* throughout wound healing. We identified a wide range of significant differences in expression of both cellular and matrix proteins, demonstrating large-scale differences in tissue composition between the two species at baseline and during wound repair. *Acomys* displayed increased production of many proteins that have previously been associated with accelerated or scarless wound healing in other models, alongside reduced expression of fibrillar collagens and other proteins which contribute to tissue tensile strength. Together, these differences from *Mus* represent potential driving factors for both the weakened skin and regenerative capacity observed in *Acomys*. This fibrillar ECM network in *Acomys* was found to be generally less soluble than that of *Mus*, correlating with differences in expression of cross-linking enzymes and potentially aiding in tissue integrity against the reduced presence of normal matrix contributors to tissue strength. We additionally observed rapid and sustained increases in markers of pro-regenerative M2 macrophages in *Acomys*, which likely support a pro-regenerative immune environment and contribute to *Acomys* regenerative capacity. Differences were also detected in matricryptic signaling peptides, with significant elevation of arrestin and endostatin-XV observed in *Acomys* independent from expression of their parent proteins. The function of these matricryptins in wound healing is not yet well understood, yet these and other insights gained from this study provide candidate matrix components which can potentially be utilized to improve future regenerative therapies.

This study significantly expands on previous proteomic analyses of *Acomys* wound healing, primarily through the use of an ECM-optimized extraction method to access repair-regulating insoluble proteins and the employment of a species-specific protein database to improve proteome coverage over previous studies. While our *Acomys*-specific protein database enabled improved peptide identification compared to the *Mus* database, limitations still exist. The transcriptome-based approach for database generation resulted in a database lacking some *Acomys* proteins due to incomplete transcript coverage. Database entries were missing or low-quality for some ECM proteins of interest, including LOX, biglycan (BGN), and fibromodulin (FMOD), preventing identification of these proteins using the *Acomys* database and subsequently limiting comparisons of these targets between *Mus* and *Acomys* (Table S7). For key proteins of interest, such as LOX (Figure 6B), this issue was addressed by searching *Acomys* data against the *Mus* database and using only peptides shared between the two organisms for comparison. Additionally, the database lacks annotated protein isoforms, which could provide deeper insights into isoform-specific expression changes during regeneration. Future construction of a more comprehensive *Acomys* protein sequence database would aid more complete characterization of the *Acomys* proteome and enhance understanding of its remarkable regenerative abilities.

Although the time points sampled here were chosen to represent as much of the wound repair timeline as feasible, additional sampling to provide greater time resolution during

early wound healing and to track wound remodeling past 4 weeks postinjury would likely reveal insights about tissue repair dynamics that were obscured in the analysis presented here. Our data indicates a robust early inflammatory and clotting response in *Acomys*, however profiling at earlier time points (e.g., 1–3 days postinjury) could better capture the dynamics of these immediate healing events. Examining earlier time points would allow quantification of ECM and immune components during initial clot formation, immune cell influx, and re-epithelialization in *Acomys* versus *Mus*. Additionally, later time points past wound closure could reveal key differences in long-term ECM remodeling and maturation. We observed high LOX levels at week 4 in *Acomys* coinciding with reduced ECM solubility, indicating tissue cross-linking may occur later to re-establish mechanical integrity. Analyzing additional late stage time points (6 weeks onward) could determine if ECM solubility in regenerated *Acomys* tissue recovers to uninjured levels during tissue maturation. Later time points could also assess how immune cell populations resolve inflammation and facilitate transition from regeneration to homeostatic tissue maintenance. Overall, expanding the time course to very early and late wound healing events could provide a more comprehensive trajectory of scarless healing in *Acomys* compared to scarring repair in *Mus*. Despite these limitations, the thorough characterization of ECM dynamics during regenerative healing in *Acomys* presented this work significantly expands our understanding of the molecular drivers of mammalian regeneration and scarless wound repair. Ultimately, these findings bring us one step closer to identifying key regulators of tissue regeneration which may hold promise for developing new therapeutic strategies to improve wound healing outcomes.

ASSOCIATED CONTENT

Data Availability Statement

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the data set identifier PXD047626.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00810>.

Volcano plot displaying all matrisome proteins significantly elevated in uninjured *Mus* (left, orange) and *Acomys* (right, blue) (Figure S1); selected significantly enriched gene ontology (GO) term annotations for the ECM protein expression clusters; chord diagrams of *Mus* vs *Acomys* correlations between cell markers and ECM protein pairs (Figure S3) (PDF)

Tabulated proteomics results displaying protein identifications by analyzed fraction (cell, sECM, and iECM fractions displayed independently) (Table S1); tabulated proteomics results displaying protein identifications as sample totals (sum of cell, sECM, and iECM fractions) (Table S2); results of statistical comparisons via linear mixed modeling to identify differences in the time-dependent expression of matrisome proteins between *Acomys* and *Mus* (Table S3); results of statistical comparisons via FDR-adjusted *t* test to identify differences in the overall expression of matrisome proteins across all time points between *Acomys* and *Mus* (Table S4); tabulated proteomics results displaying peptide identifications as sample totals (sum of cell, sECM, and iECM fractions) for COL4A1, COL4A2, COL15A1, and

COL18A1 (Table S5); tabulated proteomics results displaying protein identifications as sample totals (sum of cell, sECM, and iECM fractions) for proteins discussed in the [Comparison with Other Regenerative Models](#) section of the [Results and Discussion](#) (Table S6); list of matrisome proteins identified in previous atlas of *M. musculus* extracellular matrix composition⁴⁰ but not present in the *Acomys* protein sequence database used in this study (Table S7) (XLSX)

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

LC-MS/MS, Liquid chromatography tandem mass spectrometry; LC, Liquid chromatography; PASEF, Parallel accumulation-serial fragmentation; LFQ, Label-free quantification; FDR, False discovery rate; DB, Protein sequence database; (PTM), Post-translational modification; W0–4, Week 0–4 post injury; ECM, Extracellular matrix; FACIT, Fibrillar-associated collagens with interrupted triple helices; FASP, Filter-aided sample preparation; RNA-seq, RNA sequencing; RBB, Reciprocal best BLAST; OBB, One-way best BLAST; CHAPS, 3-((3-Cholamidopropyl) dimethylammonio)-1-propanesulfonate; EDTA, Ethylenediaminetetraacetic acid; HA, Hydroxylamine hydrochloride; Gnd-HCl, Guanidine hydrochloride; FA, Formic acid

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